

1. Introduction

This document provides operational instructions for the correct collection of biological samples by patients of Toscana Diagnostica S.r.l. Strict adherence to the instructions herein is essential to ensure the reliability of analytical results and the diagnostic validity of examinations.

Single-use containers for sample collection are provided free of charge by Toscana Diagnostica or are available at pharmacies. In case of doubt regarding preparation or collection procedures, patients are invited to contact the Collection Point or the Analysis Laboratory.

2. Venous Blood Collection

For venous blood collection the patient must:

Fast for 8-12 hours before the collection;
Avoid intense physical exertion in the 12 hours prior;
Not smoke between waking and the time of collection;
Not consume alcohol in the 12 hours prior to collection.

3. Pre-Prandial Blood Glucose

The blood draw must be performed 2 hours after breakfast.

4. Post-Prandial Blood Glucose

The blood draw must be performed 2 hours after the midday meal.

5. Glycaemic Curve and Gestational Glycaemic Curve

The patient must:

Arrive at the Collection Point at opening time;
Have fasted for at least 8 hours and no more than 14 hours before the collection.
The patient must NOT modify their diet in the 3 days preceding the examination and must maintain their usual physical activity. After performing the basal blood draw for glycaemia determination and verifying eligibility for the test, the patient will be administered the quantity of glucose solution prescribed by the physician or according to the type of curve, which must be consumed within 5 minutes. The patient must remain on-site for the entire duration prescribed by the type of curve ordered by the attending physician.

Glycaemic Curve Chart:

Gestational glycaemic curve (code 1166): 75 g glucose — Points: Basal | 60' | 120'
75 g glucose load curve (code 1079): 75 g glucose — Points: Basal | 30' | 60' | 90' | 120' | 150'
Insulinaemic curve (code 6533): 75 g glucose — Points: Basal | 30' | 60' | 90' | 120'
Note: the insulinaemic curve points may be modified at the physician's request.

6. Medications

The patient must present at the Collection Point fasting, for the blood draw before taking morning medications.

7. Complete Urinalysis Collection

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Wash external genitalia thoroughly. For complete urinalysis, collect a sample of the first morning urine, discarding the initial stream. Use single-use containers provided free of charge by Toscana Diagnostica or available at pharmacies. Women are advised to wait at least three days after the end of menstruation. Close the container tightly with the cap after collecting urine up to the MAX line of the container (50 mL).

The sample must be delivered as soon as possible after collection.

8. Urine Collection for Urine Culture

Urine must be collected in a sterile wide-mouth container. The test is performed on the first morning urine following the "midstream" technique with the following procedure:

Wash hands thoroughly;
Wash external genitalia thoroughly with soap and water, rinse;
Women must hold the labia apart with one hand and men must keep the foreskin fully retracted during cleansing and sample collection;
Discard the initial urine stream;
Collect the following urine directly into the container;
Discard the final urine stream;
Close the container quickly at the end of collection, ensuring a quantity of urine up to the MAX line of the container (50 mL);
Deliver as soon as possible; if delivery to the Laboratory is delayed, the urine sample must be stored in a refrigerator for a maximum of 24 hours.
Any antibiotic therapy must have been discontinued for at least one week unless otherwise indicated by the attending physician.

9. Urine Collection for Urine Culture in Paediatric Patients

Wash hands thoroughly with soap and water;
Thoroughly clean the child's external genitalia;
Apply the sterile collection bag (available at pharmacies) ensuring it adheres well to the skin;
Leave the bag in place for no more than 60-90 minutes; if necessary, replace it and repeat the procedure;
Once urination has occurred, remove the bag, seal it carefully, place it sealed — without transferring — into a sterile plastic container for urine culture and deliver as soon as possible to the Laboratory/Collection Point. If delivery to the Laboratory is delayed, the urine sample must be stored in a refrigerator for a maximum of 24 hours.

10. 24-Hour Urine Collection (Untreated)

The 24-hour urine collection must be performed for numerous analytes including: Aluminium, Amylasuria, Uric Acid, 5-Hydroxyindoleacetic Acid, Homovanillic Acid, Vanillylmandelic Acid, Urinary Adrenaline-Noradrenaline, Alpha-1 Microglobulin, Urinary Aldosterone, Arsenic, B2 Microglobulin, Cadmium, Urinary Calcium, Cobalt, Chromium, Kappa and Lambda Light Chains, 17-Ketosteroids, Cystine, Citrate, Urea and Creatinine Clearance, Urinary Chloride, Coproporphyrins, Urinary Cortisol, Cotinine, Creatinine, Cross-links, Urinary Dopamine, Urinary Protein Electrophoresis, Iron, Urinary Phosphorus, Glycosuria, Hydroxyprolinuria, Urinary Immunofixation, Urinary Iodine, Urinary Magnesium, Mercury, Urinary Metanephrines, Methoxytyramine, Microalbuminuria, Nickel, Urinary

Normetanephrines, Urinary C-Peptide, Lead, Total Porphyrins, Urinary Potassium, Proteinuria, Copper, Urinary Sodium, Uroporphyrins, Urea, Serotonin.

Special notes for specific analytes:

Vanillylmandelic Acid: starting 48 hours before and throughout the urine collection period, a diet free of the following foods must be observed: avocado, bananas, kiwi, dried fruit, coffee, tea, chocolate. During this period it is advisable, after consulting the physician, to avoid taking medications; otherwise, report the names of medications taken at the time of the examination.

17-Ketosteroids: starting 48 hours before and throughout the urine collection period, a diet free of the following foods must be observed: pineapple, avocado, bananas, kiwi, blackberries, red plums, aubergines, tomatoes, dried fruit, coffee, tea, chocolate. During this period it is advisable, after consulting the physician, to avoid taking medications.

24h Hydroxyprolinuria: starting 48 hours before and throughout the urine collection period, avoid the following foods: meat, meat derivatives, broth and meat extracts, products containing gelatine, fish, sweets, eggs. During this period it is advisable, after consulting the physician, to avoid taking medications.

Coproporphyrins, Total Porphyrins, Uroporphyrins: keep protected from light.

Materials required: 1 x 24h urine container (available at pharmacies) and 1 urine specimen cup (non-sterile).

Collection procedure:

Begin collection by completely emptying the bladder, discarding the first morning urine, and noting the start time (e.g. 7:00 am);

Collect all urine passed during the day and night into the container;

The following morning, finish the collection with the urine passed at the same time as the previous day (e.g. 7:00 am) in the same container;

At the end of collection, mix the 24h urine thoroughly;

Pour a quantity (min. 50 ml) into the specimen cup;

Note on the cup the total quantity collected in the 24h urine container and deliver the sample to the Laboratory.

N.B.: urine must be kept in a cool place during the entire collection period.

11. 24-Hour Urine Collection (Acidified)

The acidified 24-hour urine collection must be performed for the following analyte: 24h Urinary Oxalates.

Materials required: 1 x 24h urine container (available at pharmacies) and 1 urine specimen cup (non-sterile) to be delivered to the Laboratory.

Collection procedure:

Begin collection by completely emptying the bladder, discarding the first morning urine, and noting the start time (e.g. 7:00 am);

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Place 2 tablespoons of commercially available hydrochloric acid at 10-12% concentration at the bottom of the collection container and add urine passed during the day and night, mixing thoroughly after each addition;

The following morning, finish the collection with the urine passed at the same time as the previous day (e.g. 7:00 am) in the same container;

At the end of collection, mix the 24h urine thoroughly;

Pour a quantity (min. 50 ml) into the specimen cup;

Note on the cup the total quantity collected and deliver the sample to the Laboratory.

N.B.: urine must be kept in a cool place during the collection period.

WARNING: USE EXTREME CAUTION WHEN HANDLING HYDROCHLORIC ACID AS IT IS HIGHLY CORROSIVE AND IRRITANT.

12. Stool Collection for Stool Culture, Parasite Testing, Complete Examination

Stool must be collected with the small spoon normally provided in commercially available containers, in a quantity the size of a walnut. If delivery to the Laboratory is delayed, the sample may be stored in a refrigerator until delivery (maximum 24 hours). Please note:

For parasitological examination: it is advisable to perform the test on three fresh morning stool samples collected on different days and delivered each day;

For stool culture: any antibiotic therapy must have been discontinued for at least one week unless otherwise indicated by the attending physician; during collection, care must be taken not to contaminate the stool with toilet water or urine to avoid compromising the microbiological result. If delivery to the Laboratory is delayed, the sample may be stored in a refrigerator until delivery (maximum 24 hours);

For complete examination: a sterile container must be used, collecting any mucous or haematic portions, and delivering as soon as possible to the Laboratory. If delivery to the Laboratory is delayed, the sample may be stored in a refrigerator until delivery (maximum 24 hours).

13. Scotch Tape Test

Perform in the morning without having carried out hygiene procedures. Use transparent adhesive tape, placing the adhesive side against the anal opening and applying light pressure for 30 seconds. Spread the tape on the surface of the glass slide previously provided by the Laboratory, avoiding creases. Collection must be performed on the same day as delivery.

Enterobius Vermicularis eggs are sticky and highly infectious; therefore, it is recommended to wear protective gloves during all collection phases.

14. Stool Collection for Faecal Occult Blood and Faecal Calprotectin

Stool for faecal occult blood and calprotectin testing must be collected using the container provided by the Laboratory, following the enclosed instructions. Please note:

Keep the container out of the reach of children;

The liquid inside the container must not be ingested and must not come into contact with eyes or mucous membranes. The tube has been specifically designed to prevent liquid leakage. Contact a physician immediately if contact has occurred;

Collect the sample immediately after defecation;

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If requested, note on the container: Collection Date (D), Sample Identification Code (ID), Patient Name (N);

Do not insert the container or any part of it into the anus;

Do not collect stool in case of diarrhoea, during menstruation, in case of rectal bleeding (e.g. haemorrhoids or other causes) or if contaminated by urine;

Use the device only once;

Do not open or attempt to open the transparent cap;

If using a support for sample collection, follow the instructions on the support. Then follow the sample collection procedure from step 3.

Collection instructions:

Place toilet paper inside the toilet bowl;

Defecate onto the toilet paper. If possible, sit facing backwards;

Unscrew the green cap fitted with the sampling stick;

Insert the sampling stick into four different points and scrape in a cross pattern on the surface. The grooves must be covered;

Replace and tightly screw the green cap with the sampling stick. Do not reopen;

Store the collection device in the refrigerator and in the dark. Return the device as soon as possible for analysis.

15. Stool Collection: Faecal Elastase

Stool for faecal elastase testing must be collected using the container provided by the Laboratory or available at pharmacies. Please note:

Keep the container out of the reach of children;

Collect the sample immediately after defecation;

Do not collect stool in case of diarrhoea, during menstruation, in case of rectal bleeding (e.g. haemorrhoids or other causes) or if contaminated by urine;

Use the device only once.

Collection instructions:

Place toilet paper inside the toilet bowl;

Defecate onto the toilet paper. If possible, sit facing backwards;

Unscrew the blue cap and use the spatula provided in the container to collect a small amount of stool.

Do not fill the container;

Screw the cap back on carefully and store in the refrigerator;

Deliver the collected sample to the Collection Point within the same day.

16. Vaginal and Vagino-Rectal Swab

The patient (female):

Must abstain from sexual intercourse for 24 hours before the examination;

Must have discontinued any antibiotic and antifungal therapy for at least one week;

Must not have performed vaginal douching in the 24 hours before the examination;

The test cannot be performed during menstruation (except where acute symptoms make the test indispensable during this period).

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For the vaginal swab specifically, intended for culture, a fresh wet mount must reach the laboratory, performed at the time of collection, accompanied by the appropriate completed form. For the vagino-rectal swab, no wet mount is required.

17. Cervical and Endocervical Swab

The patient (female):

Must abstain from sexual intercourse for 24 hours before the examination;
Must have discontinued all antibiotic therapy for at least one week and must not have performed vaginal douching in the 24 hours before the examination;
The test cannot be performed during menstruation (except where acute symptoms make the test indispensable during this period).

18. Urethral Swab

Female patient:

Must abstain from sexual intercourse for 24 hours before the examination;
Must have discontinued any antibiotic and antifungal therapy for at least one week;
Must not have performed vaginal douching in the 24 hours before the examination;
Must not be menstruating;
Must not have urinated in the previous three hours.

Male patient:

Must abstain from sexual intercourse for 24 hours before the examination;
Must have discontinued any antibiotic and antifungal therapy for at least one week;
Must not have urinated in the previous 3 hours.

19. Sputum Collection for Culture Examination

Collect in the morning on an EMPTY STOMACH in a sterile wide-mouth container; rinse the oral cavity thoroughly with water gargling; perform a deep expectoration collecting the sputum directly into the sterile container. It is important that the sample does not consist of saliva and it is essential to deliver to the Collection Point within one hour of collection.

20. Sputum Collection for Three-Sample Cytological Examination

The following instructions must be followed scrupulously:

Upon waking, rinse the oral cavity with water;
Cough deeply;
Collect the sputum directly into the container;
Rinse the oral cavity thoroughly with water gargling;
Perform a deep expectoration (saliva must not be collected);
Collect the sputum directly into the container provided by the laboratory;
Repeat the entire procedure for three consecutive days, storing the containers in a cool place.

WARNING: the solution inside the container serves as a PRESERVATIVE; therefore, it must not be discarded or dispersed into the environment. Some components of the preservative in pure form are

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toxic or flammable. However, they are present in negligible concentrations in the final solution, conferring characteristics of NON-TOXICITY and NON-FLAMMABILITY. It is nevertheless recommended to: keep containers sealed, at room temperature (15-25°C) away from heat sources and flames, do not ingest, avoid contact with skin and eyes and keep out of reach of children. In case of eye contact, wash thoroughly with water and decongestant eye solution. In case of ingestion, drink plenty of water, avoid emetics, milk or charcoal, consult a physician presenting this document. The three samples must be delivered to the Laboratory, accompanied by documentation regarding: symptoms, previous interventions, histological, cytological, radiological, ultrasound examinations of the respiratory tract and therapies performed.

21. Cervico-Vaginal Material Collection for PAP Test

The patient (female):

Must abstain from sexual intercourse for 48 hours before the examination;

Must not have performed topical therapies or vaginal douching in the 24 hours before the examination;

The test may be performed up to 5 days before the menstrual period and only 5 days after its end.

For semen analysis and for biological samples not expressly mentioned, please contact the Laboratory.

22. Sperm Culture

The patient must abstain from sexual intercourse for 24 hours before the examination; must have ceased any antibiotic therapy for at least one week; the collection must be preceded by thorough cleaning of external genitalia and hands; semen collection must occur through masturbation directly into a sterile container. In case of delayed processing, the material must be stored in a refrigerator for a maximum of 24 hours.

23. Three-Sample Urinary Cytological Examination

The following instructions must be followed scrupulously:

Discard the first morning urine;

Drink approximately one litre of water or other non-alcoholic, non-carbonated beverage;

Wash genitalia thoroughly;

After drinking, collect in the container provided by the laboratory;

Repeat the entire procedure for three consecutive days, storing the containers in a cool place.

At the end of the 3-day collection, deliver the 3 samples and the admission form properly completed in each section to the laboratory. Accurate collection of clinical information is essential for a correct cytological diagnosis.

WARNING: the solution inside the container serves as a PRESERVATIVE; therefore, it must not be discarded or dispersed into the environment. Some components of the preservative in pure form are toxic or flammable. It is nevertheless recommended to: keep containers sealed, at room temperature (15-25°C) away from heat sources and flames, do not ingest, avoid contact with skin and eyes and keep out of reach of children. In case of eye contact, wash thoroughly with water and decongestant eye solution. In case of ingestion, drink plenty of water, avoid emetics, milk or charcoal, consult a physician presenting this document.

24. Sputum for Neoplastic Cell Collection

Sputum must be collected in the morning on an empty stomach.

25. Breath Test

25.1 What is the Breath Test

The Breath Test is a non-invasive examination used for the diagnosis of food intolerances (lactose, glucose, sorbitol), or when bacterial contamination is suspected (lactulose). The test identifies alterations of the gastrointestinal system and is aimed at those seeking answers to symptoms that are often difficult to interpret such as trapped wind, diarrhoea, slow or poor digestion, bloating, constipation, flatulence and indigestion.

The Urea Breath Test is an investigation performed to diagnose gastroduodenal infection by *Helicobacter Pylori*.

25.2 Types of Breath Test

LACTOSE Breath Test: The H₂ Breath Test is indicated for determining lactose malabsorption. Lactose intolerance is the most common form of carbohydrate malabsorption affecting people of all ages. The cause is a deficiency of lactase, normally present in the small intestine. Under normal conditions, lactose is broken down by intestinal lactases into glucose and galactose which are absorbed at the jejunal-ileal level; in cases of lactase deficiency, lactose reaches the colon unaltered where it is fermented by bacterial flora producing various gases including hydrogen, methane and carbon dioxide, which partly enter the bloodstream and are eliminated through exhaled air.

LACTULOSE Breath Test: The H₂ Lactulose Breath Test is used for diagnosing alterations in Oro-caecal Transit Time (OCTT) and SIBO. The measurement of OCTT assesses the time food takes to reach the intestine, which may be reduced in cases of hypermotility diarrhoea or increased in cases of dyspeptic symptoms with constipation or slow gastric emptying.

GLUCOSE Breath Test: The H₂ Breath Test is indicated for diagnosing Small Intestinal Bacterial Overgrowth (SIBO). Bacterial overgrowth syndrome is very common, occurring in many organic pathologies but also frequently in Irritable Bowel Syndrome (IBS) or in patients chronically taking medications (antacids, proton pump inhibitors) or who have undergone abdominal surgery or in advanced age.

FRUCTOSE Breath Test: The H₂ Breath Test is indicated for diagnosing fructose malabsorption. Fructose is a naturally occurring sugar found in certain vegetables, fruits and cereals. The fructose breath test is a non-invasive method for determining fructose intolerance which manifests with diarrhoea, flatulence and abdominal pain.

HELICOBACTER PYLORI Breath Test: The test is indicated for evaluating the presence of *Helicobacter pylori* in the gastric mucosa. *Helicobacter Pylori* is a bacterium that finds its ideal growth and replication environment inside the human stomach. This microorganism is considered the only one capable of withstanding gastric acidity; this characteristic is conferred by its marked ability to produce urease, an enzyme that breaks down urea into carbonic acid and ammonia. In this way, urease neutralises gastric acids creating a microenvironment favourable to bacterial replication.

25.3 Preparation for Lactose, Lactulose, Glucose, Sorbitol, Fructose Breath Test

Preparation protocol:

The test must be performed fasting;

In the 4-5 days preceding the test, the patient must not take antibiotics, probiotics and laxatives (20 days for antibiotics);

The physician must ensure the patient does not have significant ongoing diarrhoea;

The day before the test, the patient must eat exclusively: BREAKFAST: a cup of tea; LUNCH: a plate of boiled rice with a little oil; DINNER: grilled/boiled chicken/turkey breast or boiled fish;

From 9:00 pm the day before, the patient MUST observe complete fasting (may drink still water freely);

From 7:00 am on the day of the test, the patient MUST remain fasting, not brush teeth, not smoke, and not exercise;

Rinsing the oral cavity with 0.2% chlorhexidine is recommended in the 30 minutes before the basal collection;

During the test, smoking, drinking, eating, chewing gum or sweets, or performing strenuous physical activity is not permitted.

25.4 Preparation for Urea Breath Test (Helicobacter Pylori)

To ensure adequate test accuracy, the following instructions must be followed:

Antibiotics and proton pump inhibitors (PPIs) must not have been taken in the last 4 weeks;

Gastroprotective agents must not be taken in the week preceding the test;

Fasting must be observed on the morning of the test; only still mineral water may be consumed 2 hours before the test;

The day before the test, the following diet must be followed: BREAKFAST: barley coffee, tea, dry biscuits or toast; LUNCH: rice or pasta with oil or Parmesan; AFTERNOON: barley coffee, tea, dry biscuits or toast; DINNER: white meat (chicken, rabbit or turkey), fish, rice with oil;

Do not brush teeth;

Smokers must abstain from smoking from at least the evening before.